

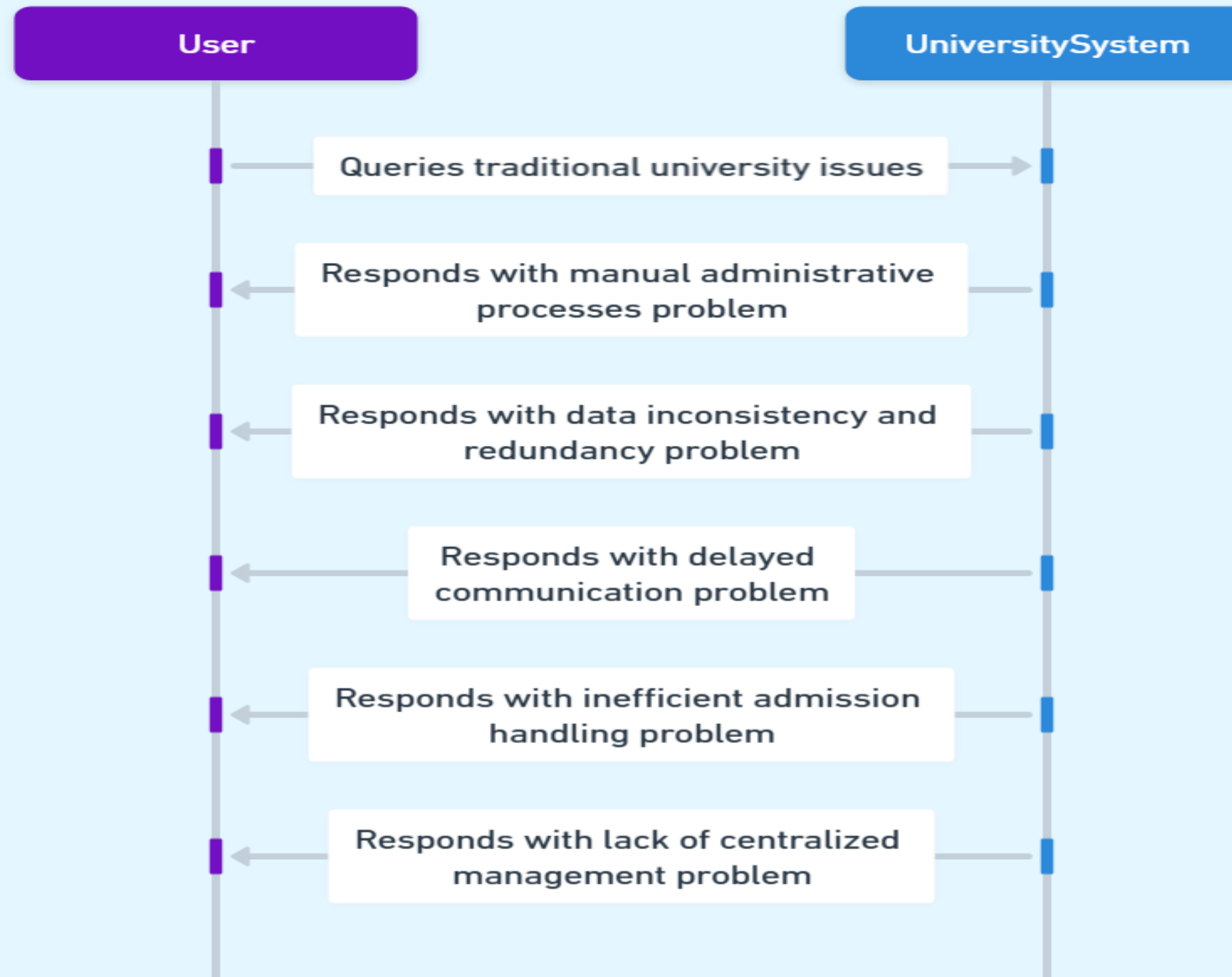
Industrial Training Defense

“University Management System: A Digital Ecosystem for Academic Excellence”

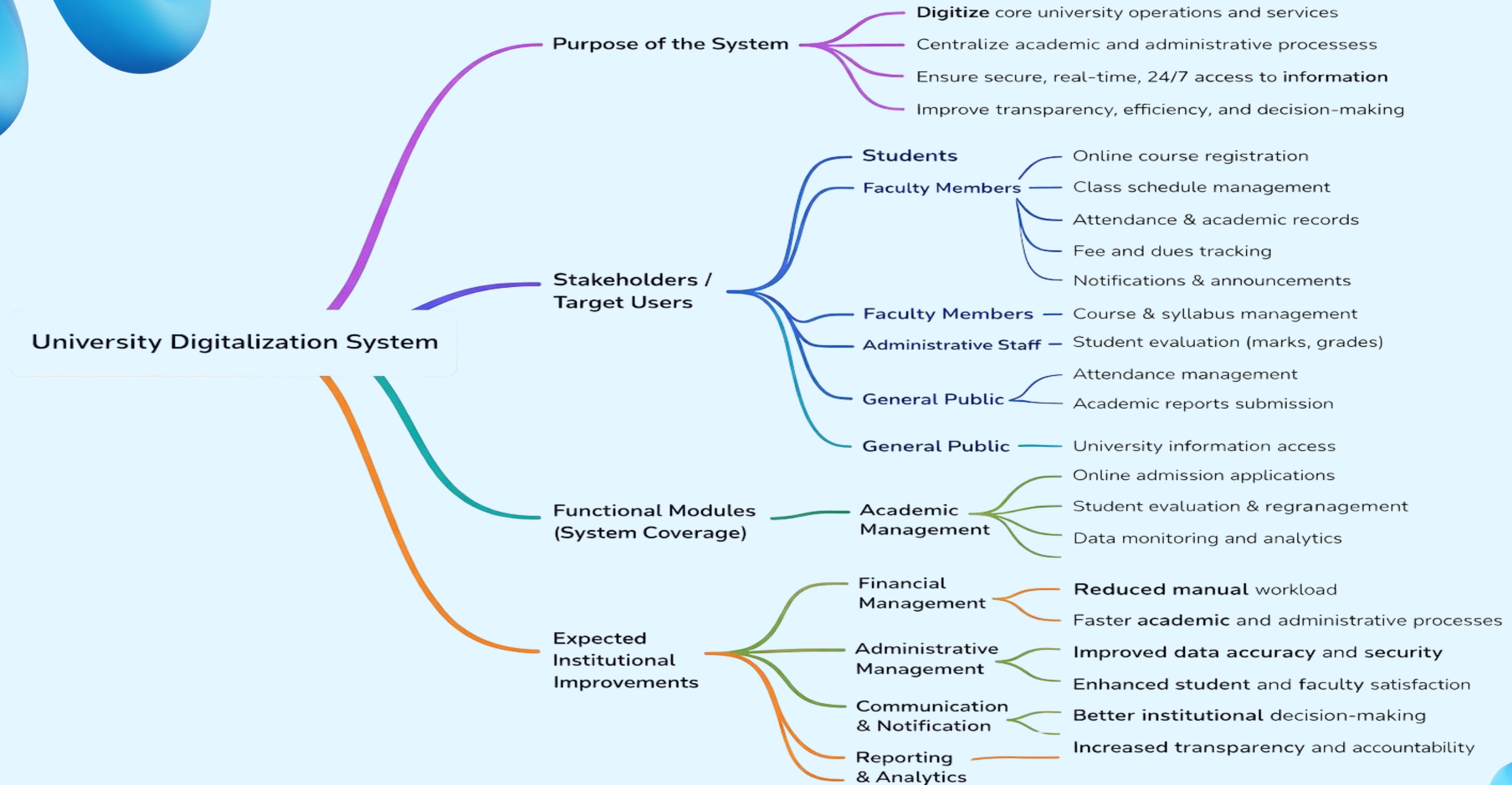
Presented by:
Md. Mahfujul Karim Sheikh
ID:11220120735
Section:7B
Northern University Of Business And
Technology Khulna

Supervised by:
Md. Mossadek Touhid
Lecturer
Department of
Northern University Of Business And
Technology Khulna

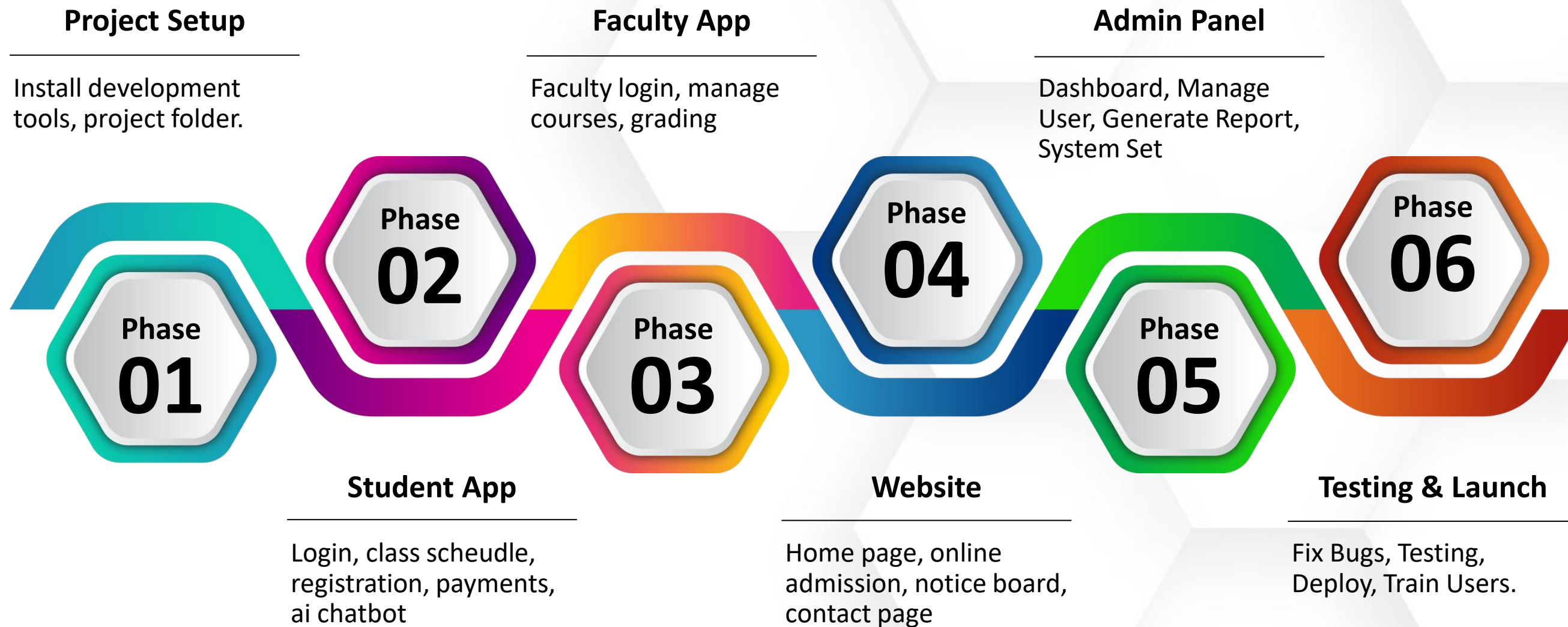
Problem Figure Out



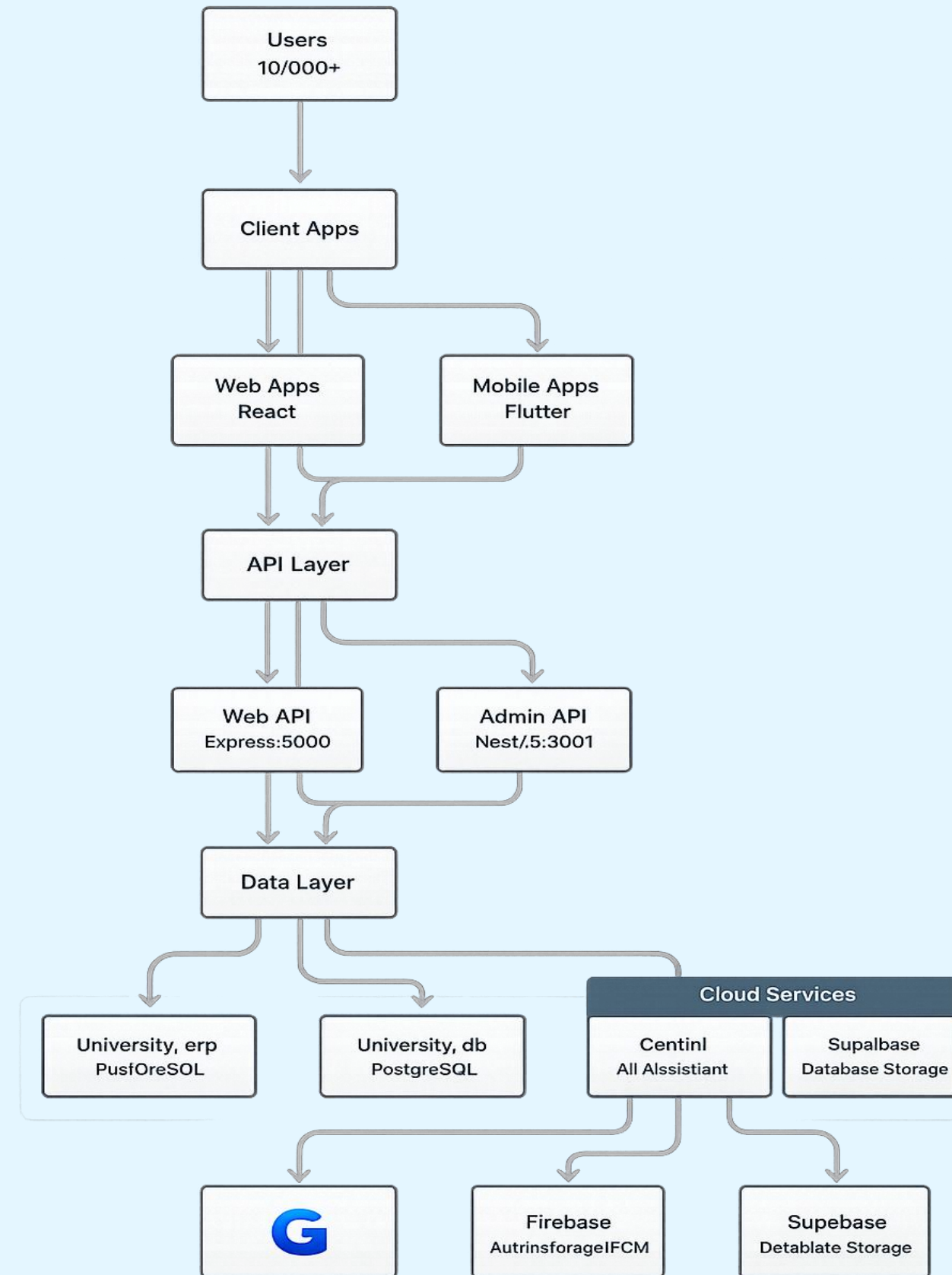
Objectives & Scope



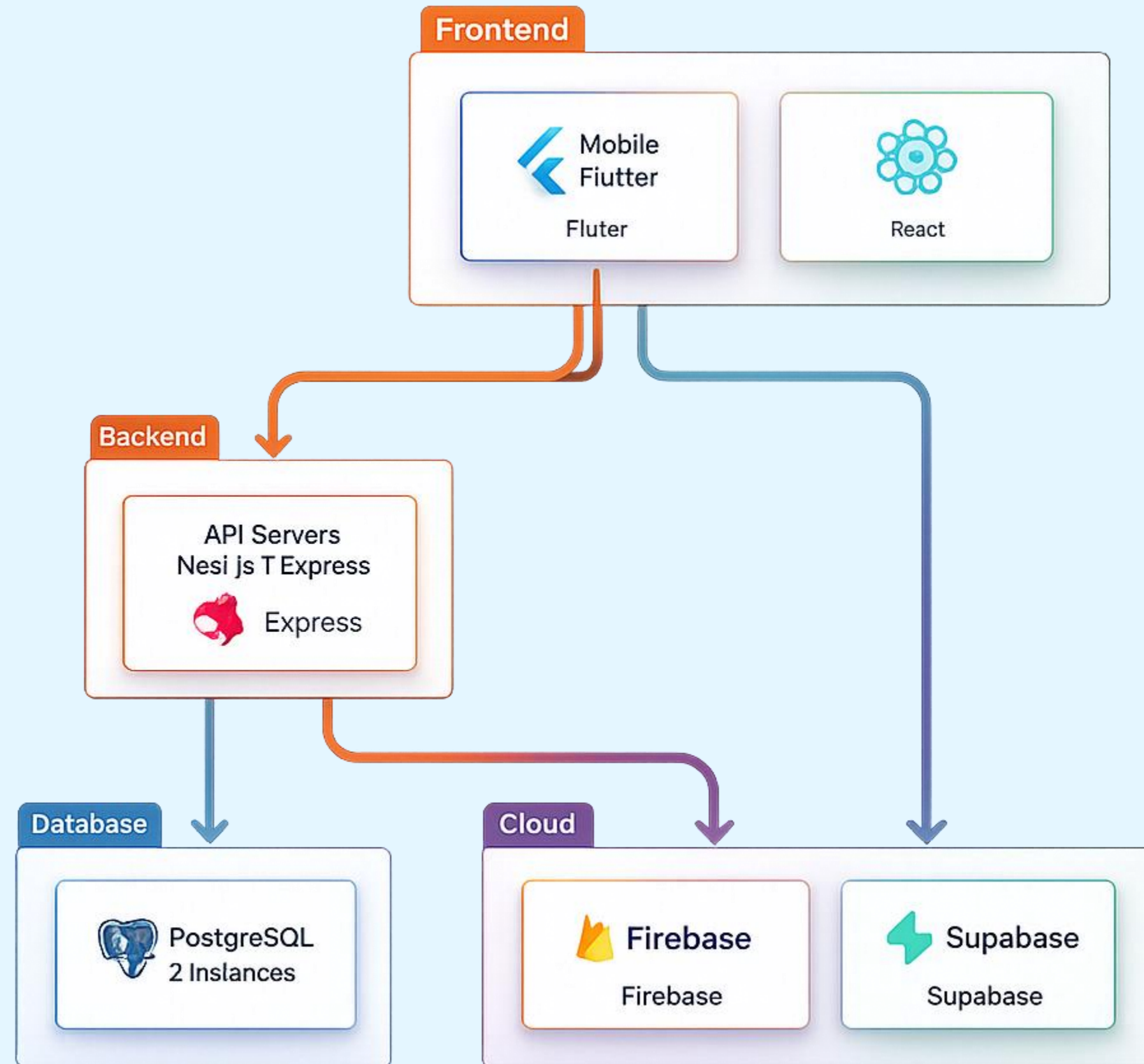
Implementation Step



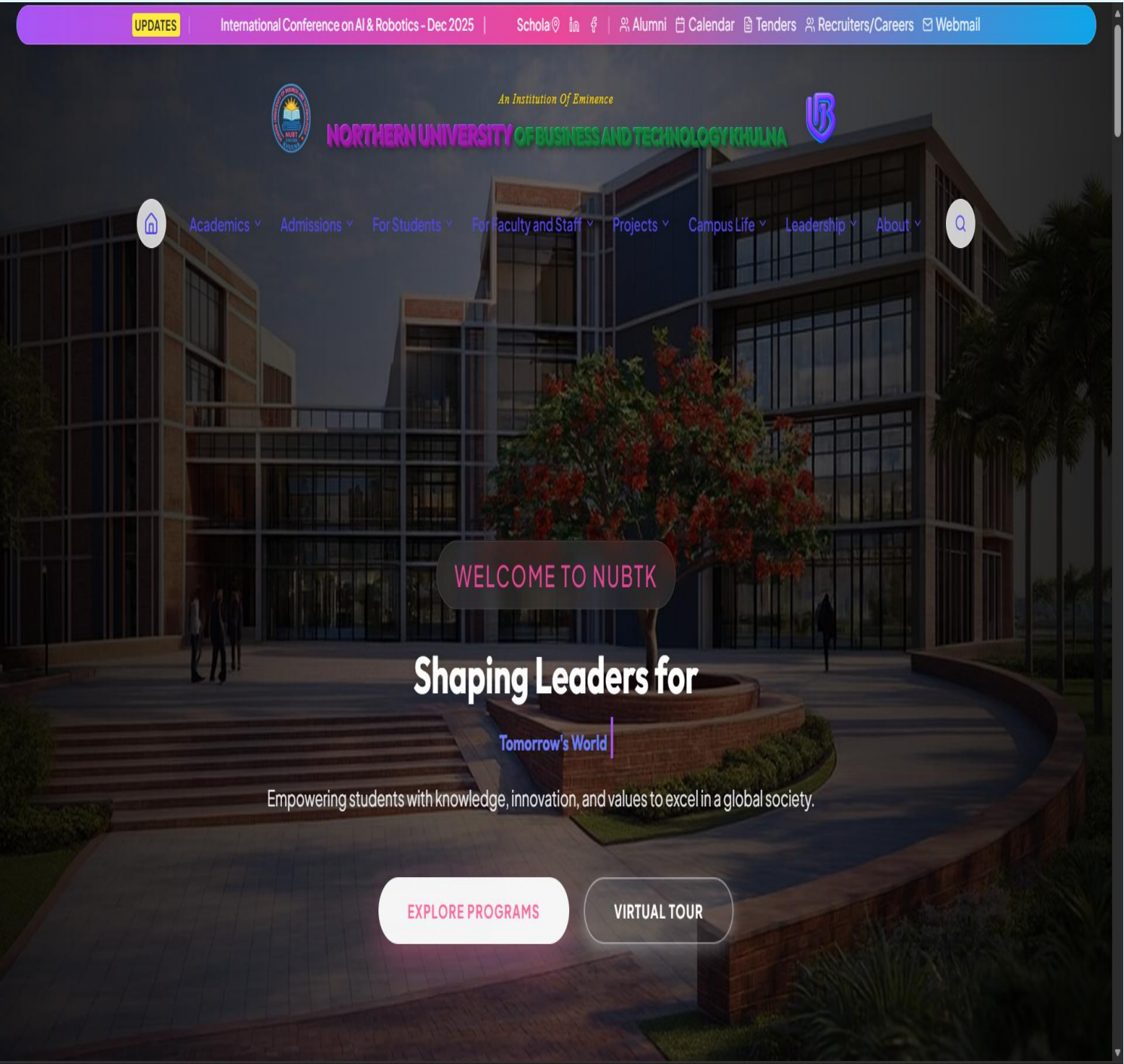
Entire System Architecture



Technology Stack



University Website



Academic System

Comprehensive overview of our academic framework and processes

Overview

Our academic system is designed to provide students with a comprehensive and rigorous educational experience. We follow a semester-based system with two main semesters per academic year, along with summer courses for additional learning opportunities.

Academic Structure

Semester System

The academic year is divided into two semesters: Fall (September to January) and Spring (February to June), each lasting approximately 16 weeks including examination periods.

Credit Hours

Courses are measured in credit hours, typically ranging from 1 to 4 credits depending on the course load and contact hours per week.

Grading System

We use a letter grading system (A+, A, A-, B+, B, B-, C+, C, C-, D+, D, F) with corresponding grade point values to calculate cumulative GPA.

Degree Requirements

Undergraduate programs typically require 120-130 credit hours for completion, including general education, major requirements, and electives.

Academic Policies

Attendance Policy: Students are expected to attend at least 80% of classes to be eligible for examinations.

Make-up Policy: Make-up exams are permitted only for valid documented reasons such as illness or family emergencies.

Late Submission: Late assignments may be accepted with penalty unless prior arrangements have been made.

Academic Integrity: Plagiarism and cheating are strictly prohibited and subject to disciplinary action.

Support Services

Academic Advising

Each student is assigned an academic advisor who provides guidance on course selection, degree planning, and career development.

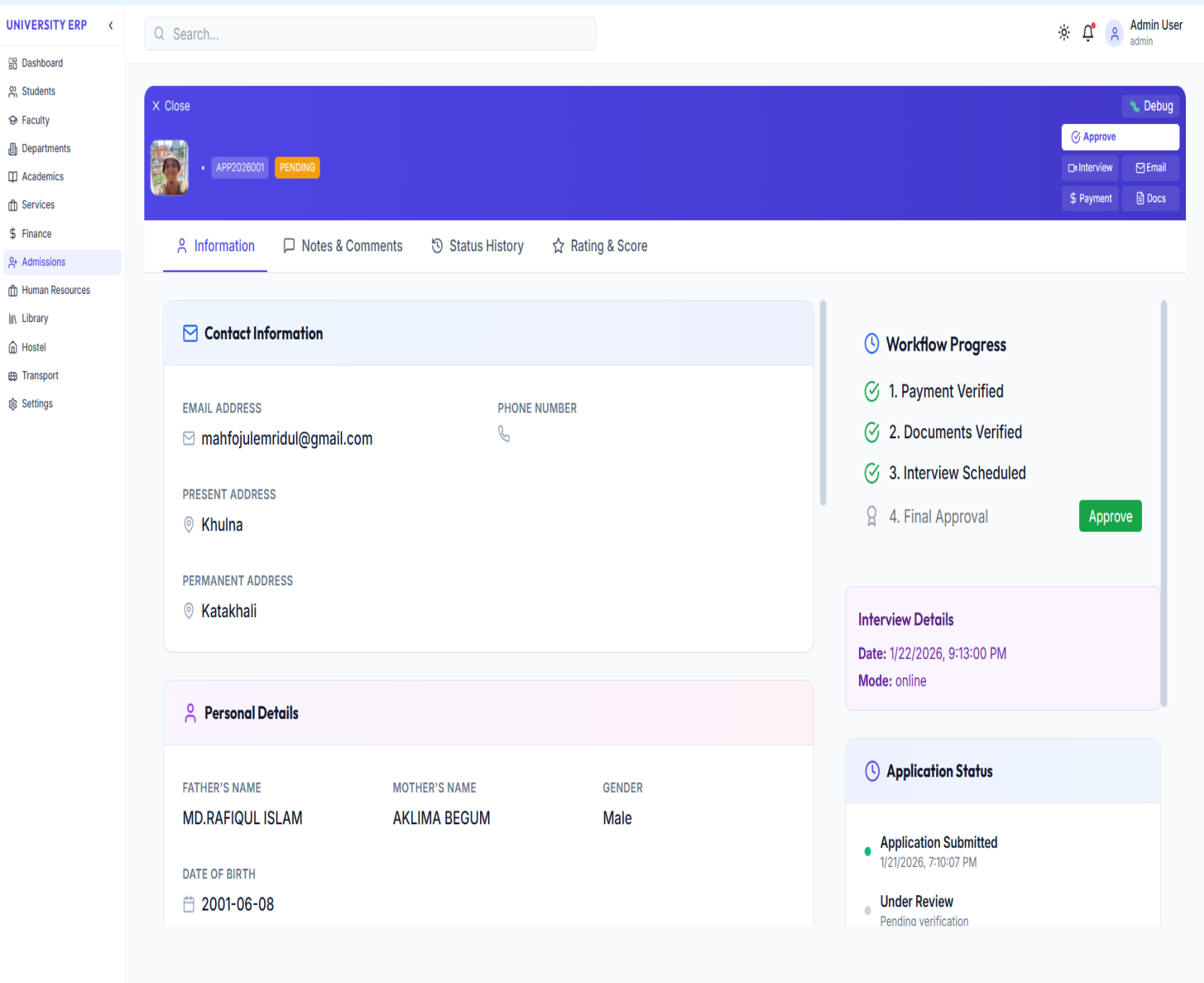
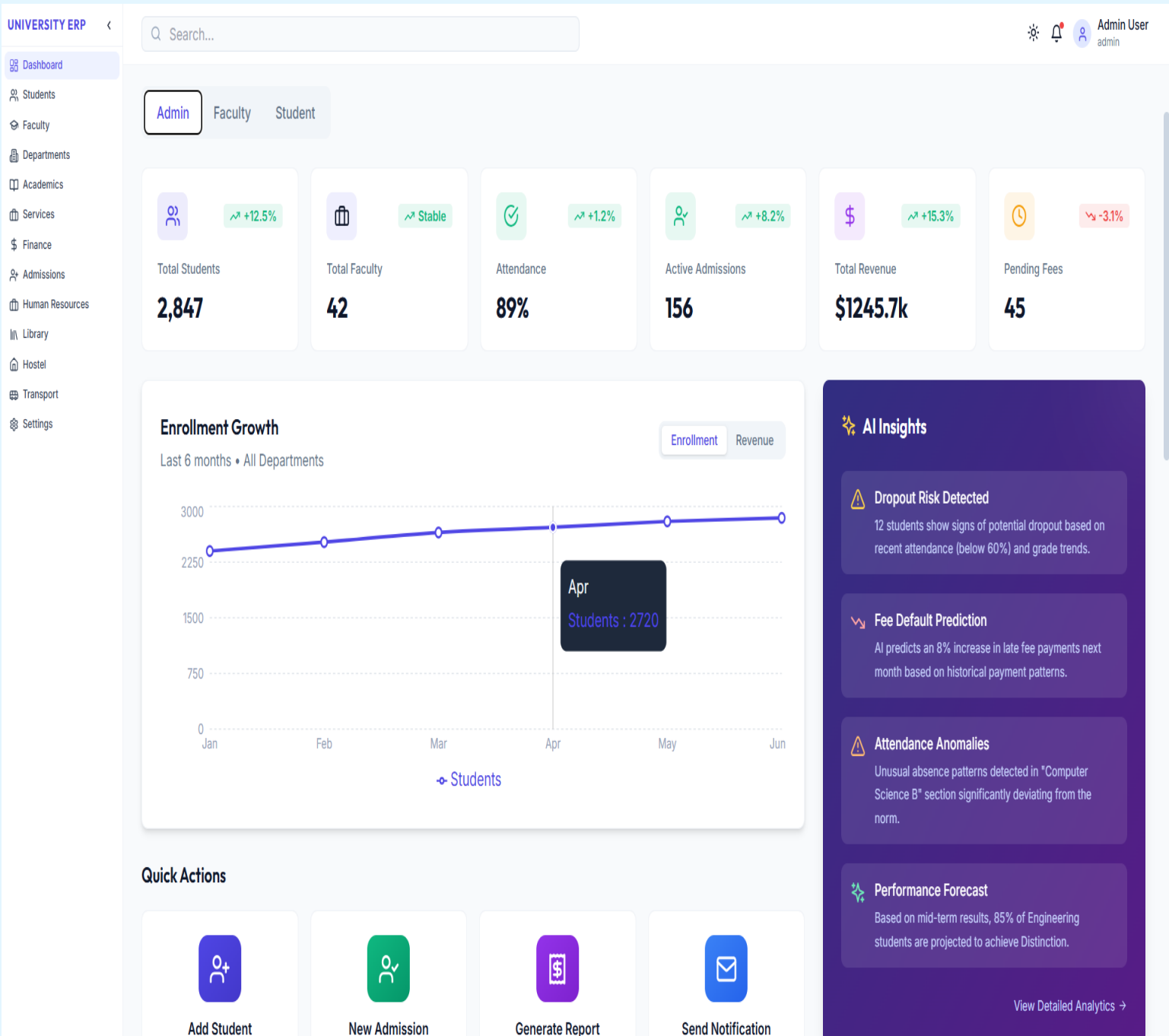
Tutoring Services

Free tutoring services are available for challenging subjects through our Student Success Center.

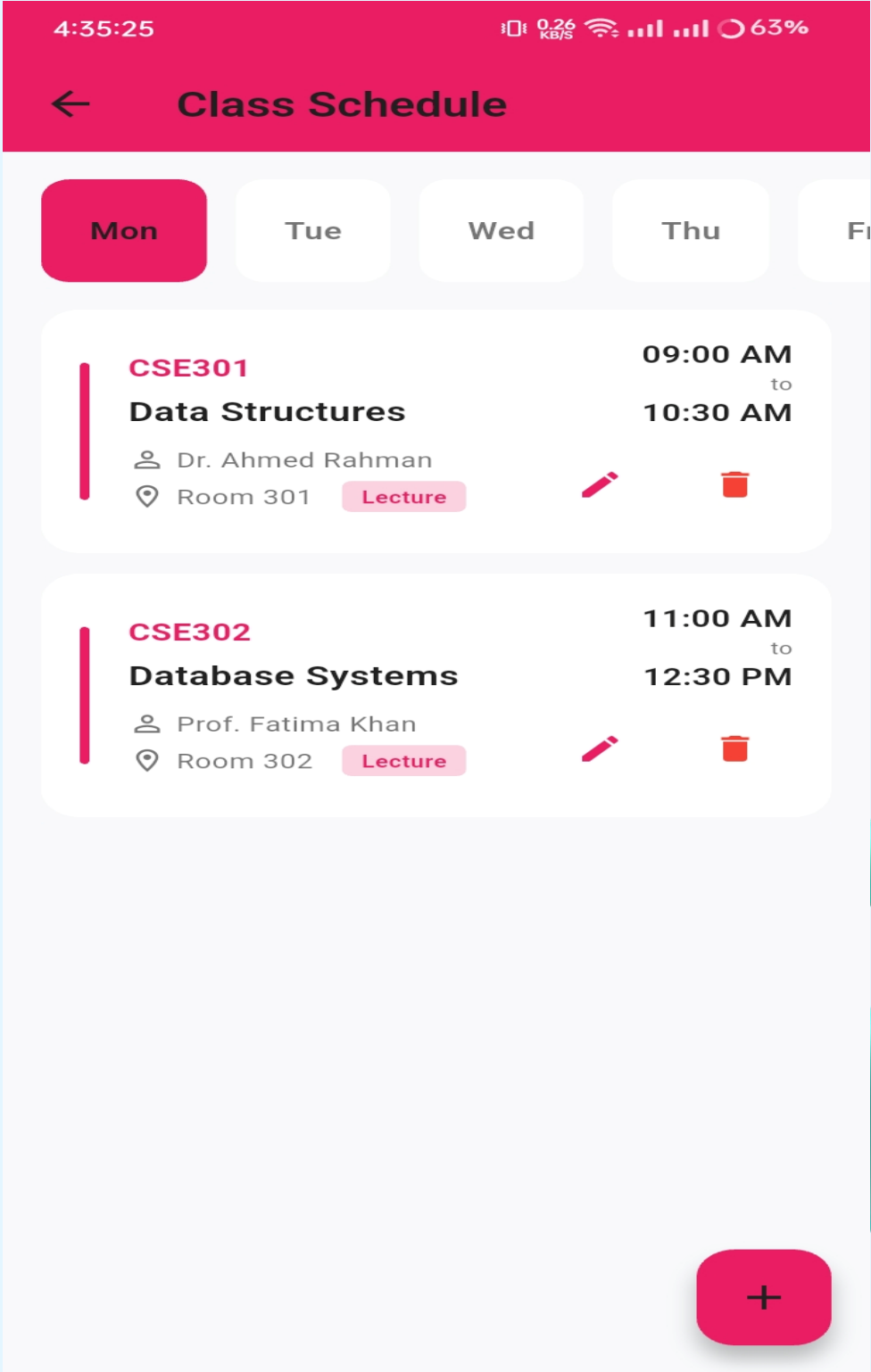
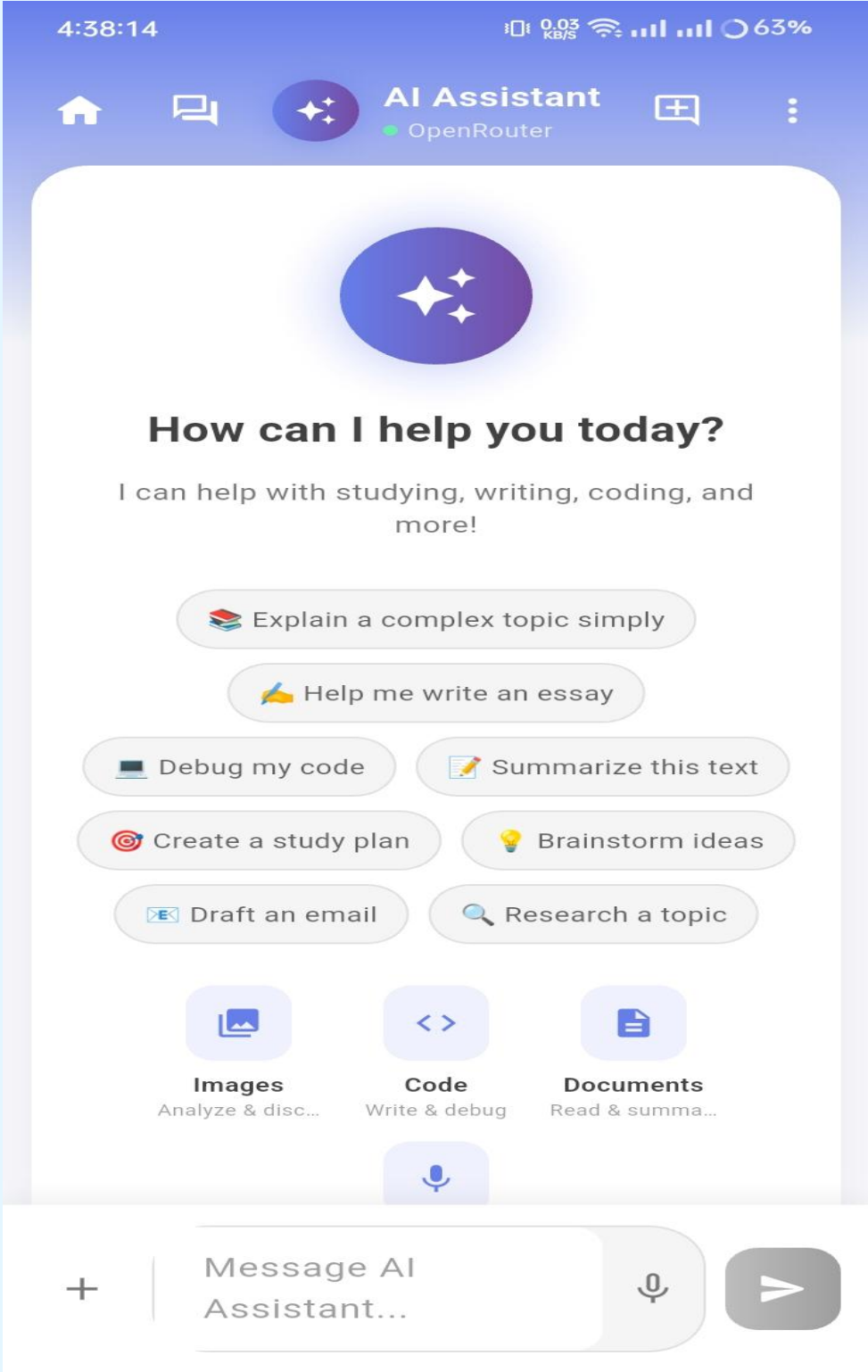
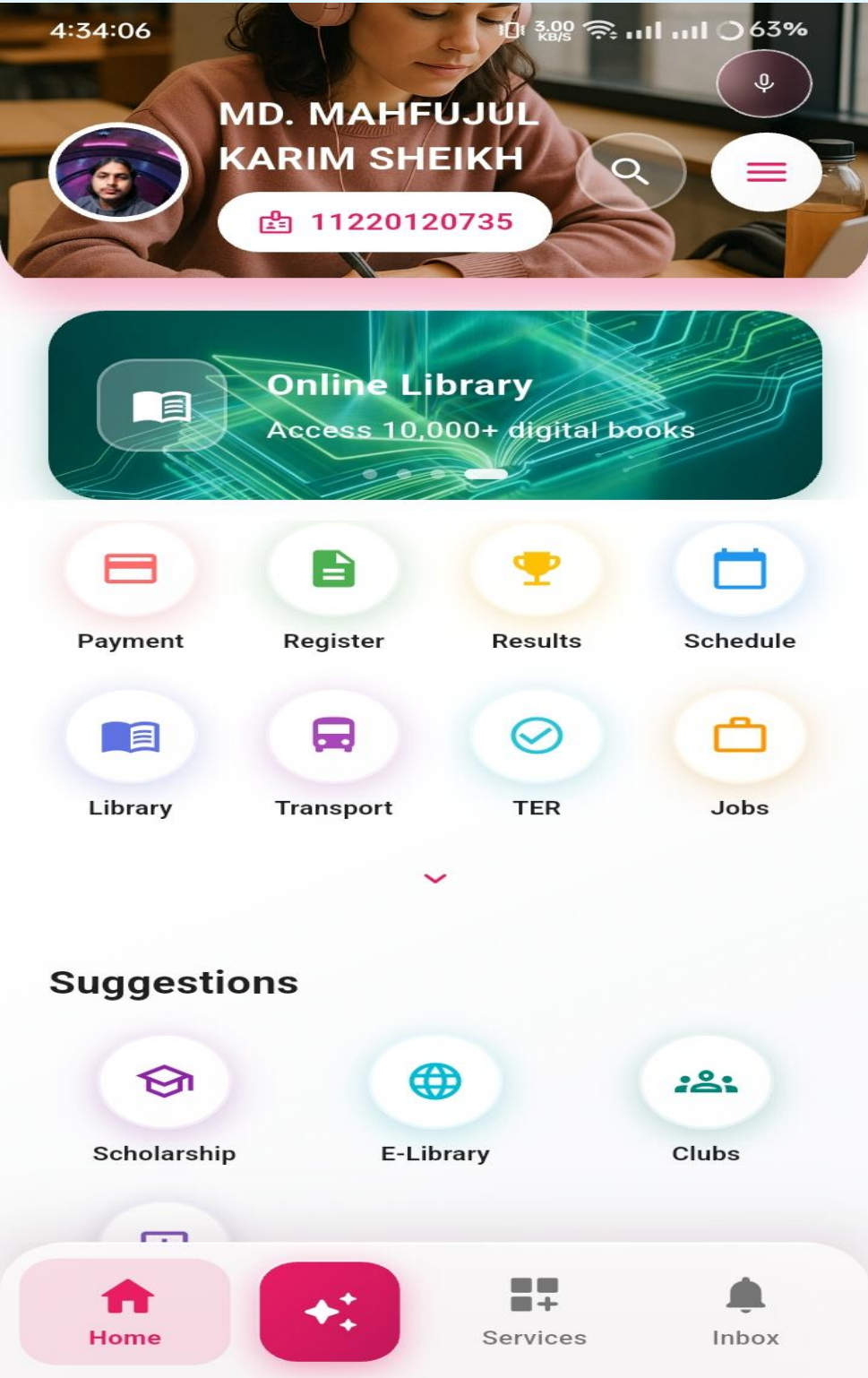
Library Resources

Our modern library offers extensive digital and physical resources, study spaces, and research assistance.

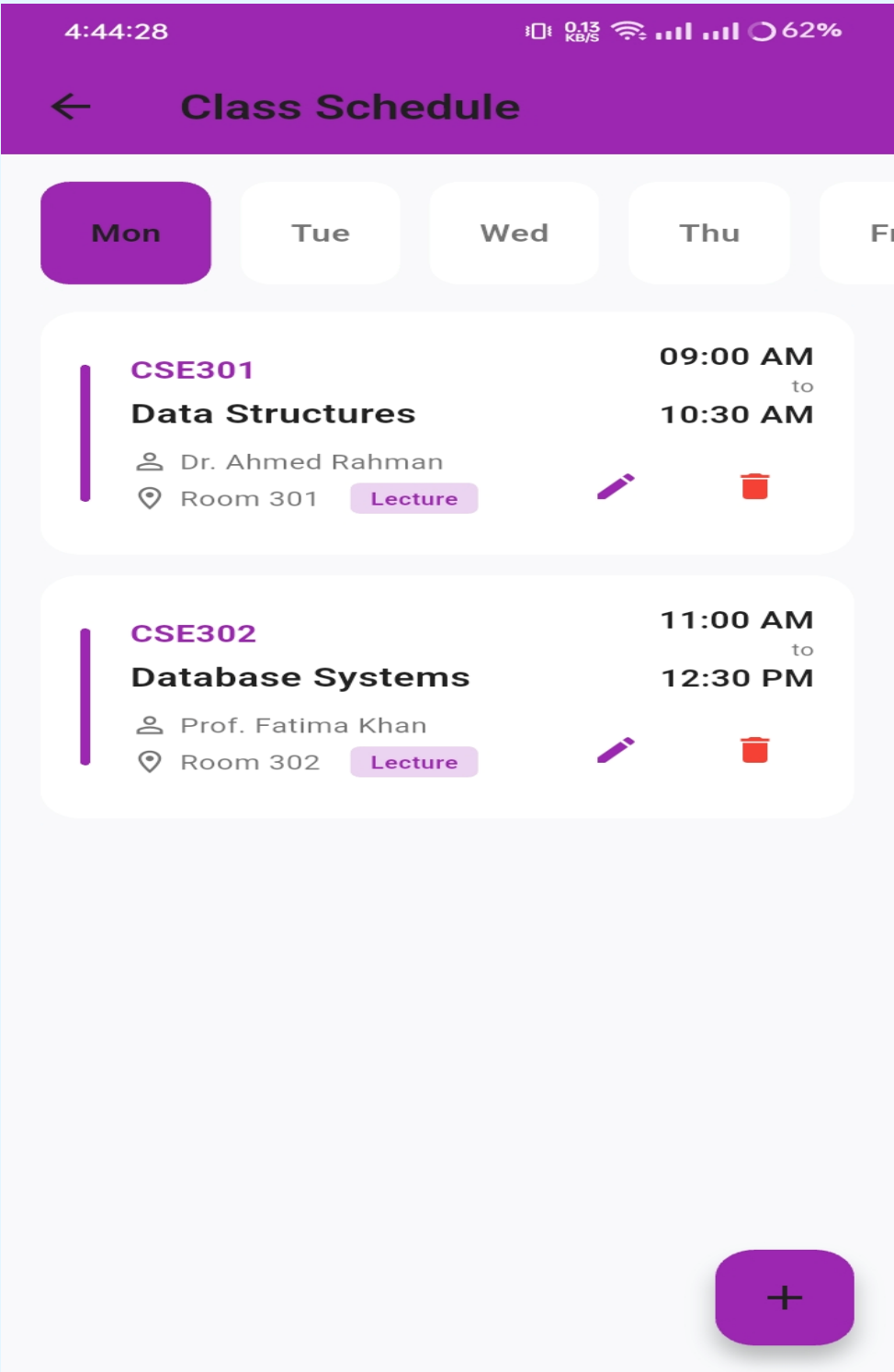
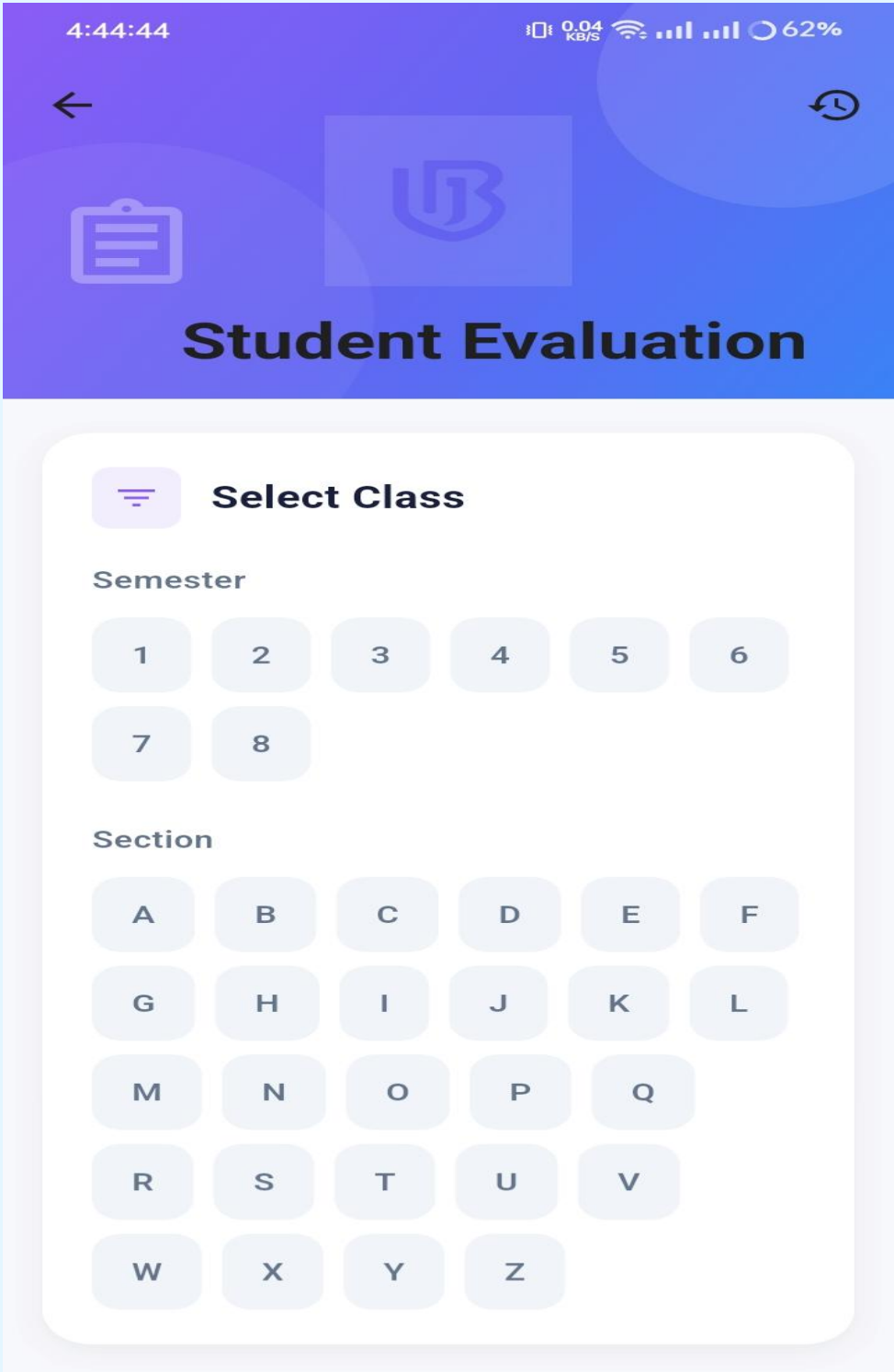
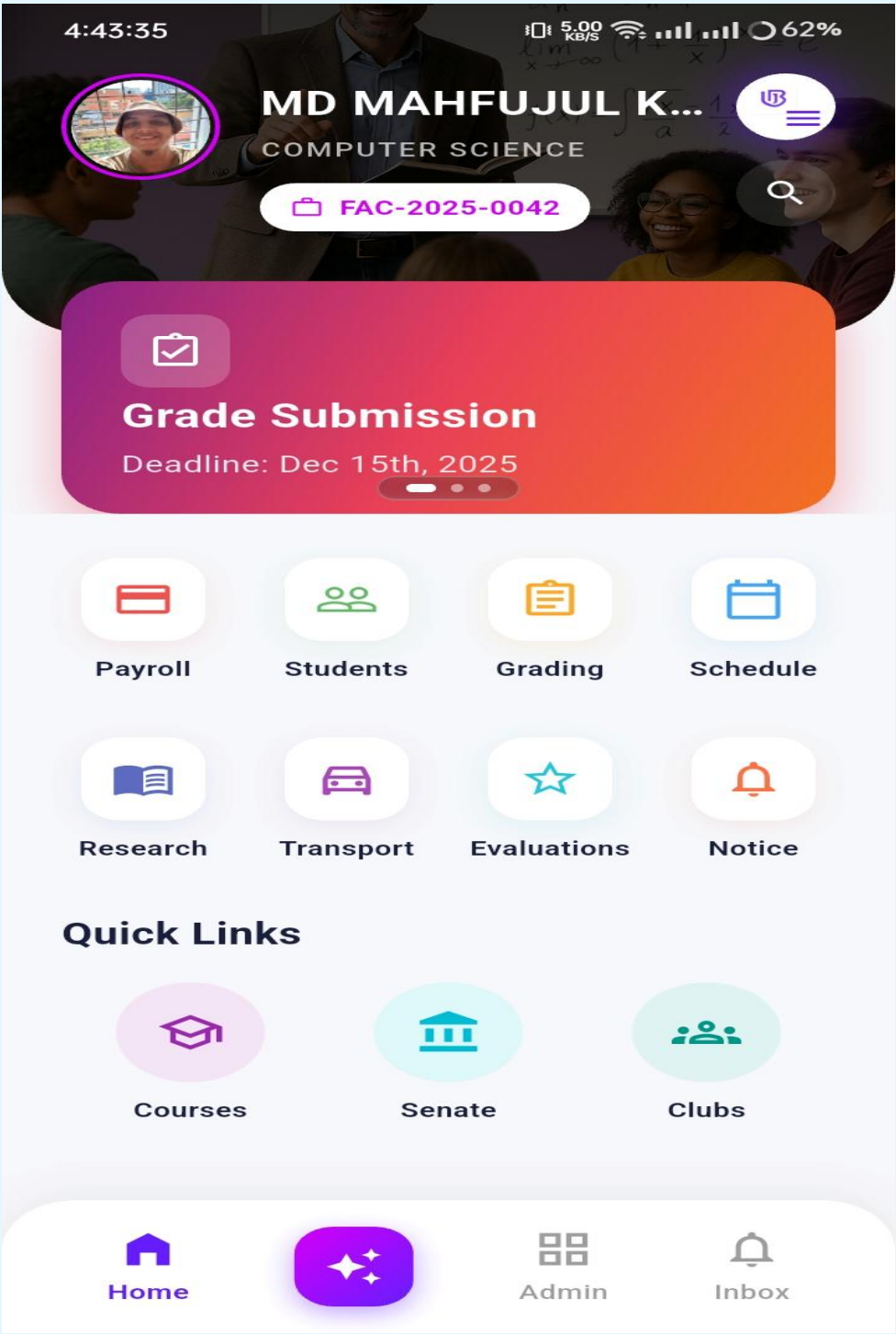
Admin Panel



Student App



Faculty App



Security & Authentication



JWT-Based Authentication

- ▶ Stateless token-based authentication
- ▶ Secure session management
- ▶ Token expiration and refresh mechanisms



Role Based Access Control (RBAC)

- ▶ Granular permission management
- ▶ Role hierarchy: Admin → Faculty → Student
- ▶ Protected routes and endpoints



Firebase Authentication

- ▶ Multi factor authentication Support
- ▶ Social login integration (Google etc)
- ▶ Secure credential storage



Data Security Measures

- ▶ Input validation and sanitization
- ▶ SQL Injection prevention
- ▶ Encrypted sensitive data storage



Key Features



Ordine Admission System

- ▶ End to end digital admission process
- ▶ Document verification and tracking



Student Academic Dashboard

- ▶ Real-time grade tracking and GPA calculation.
- ▶ Course materials and schedule access



Faculty Attendance Management

- ▶ Digital attendance marking system
- ▶ Automated reporting and analytics



Notices and Class Schedules

- ▶ Real time notifications and updates
- ▶ Centralized information distribution



Notires and Class Schedules

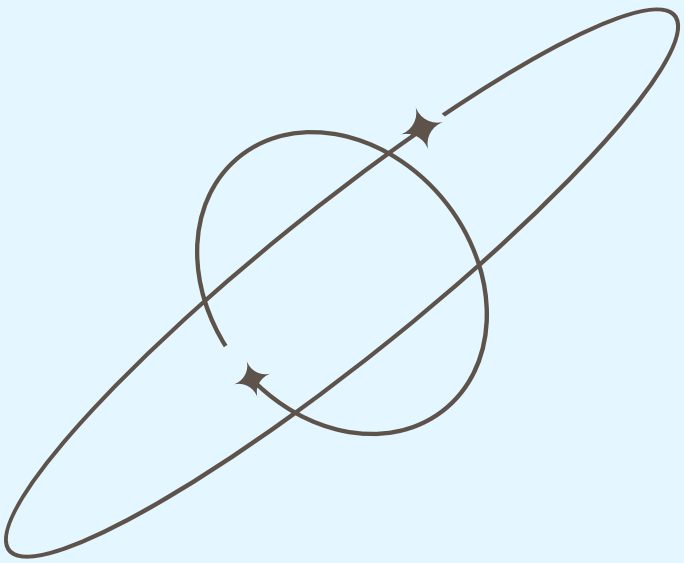
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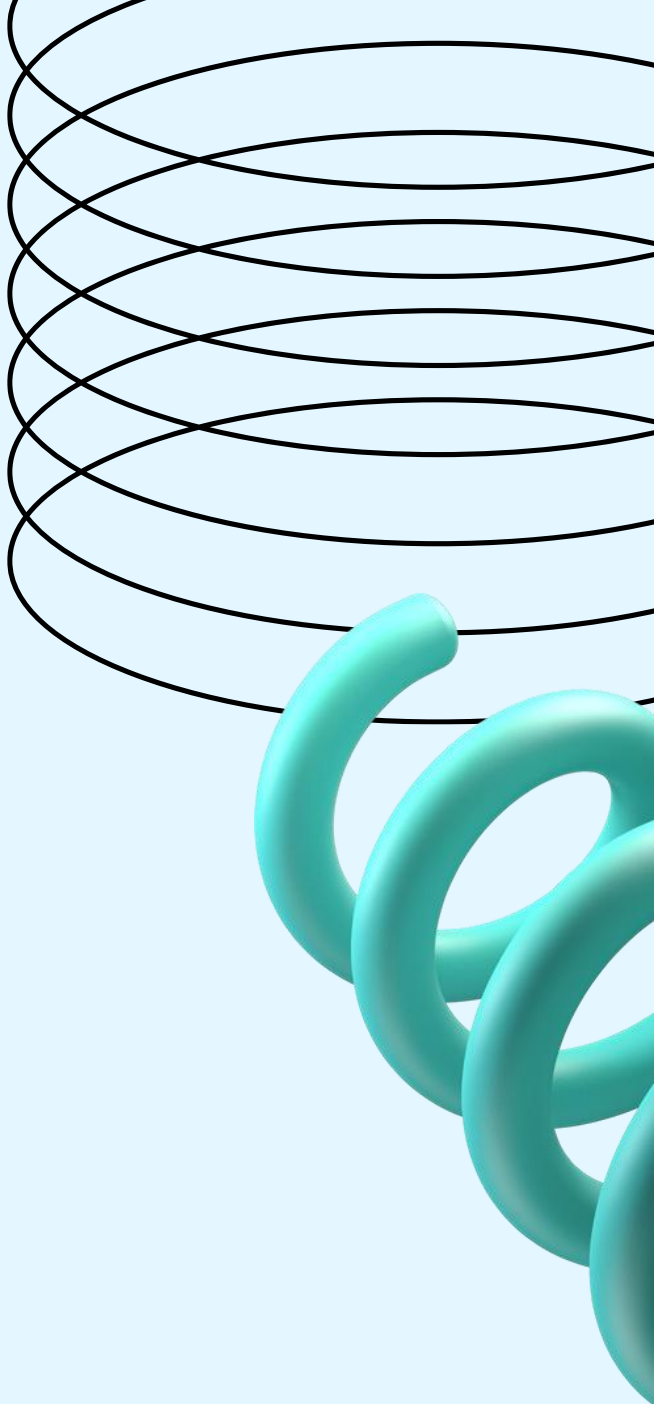
AI-Powered Academic Assistant

- ▶ Intelligent query resolution
- ▶ Course recommendations at guidance





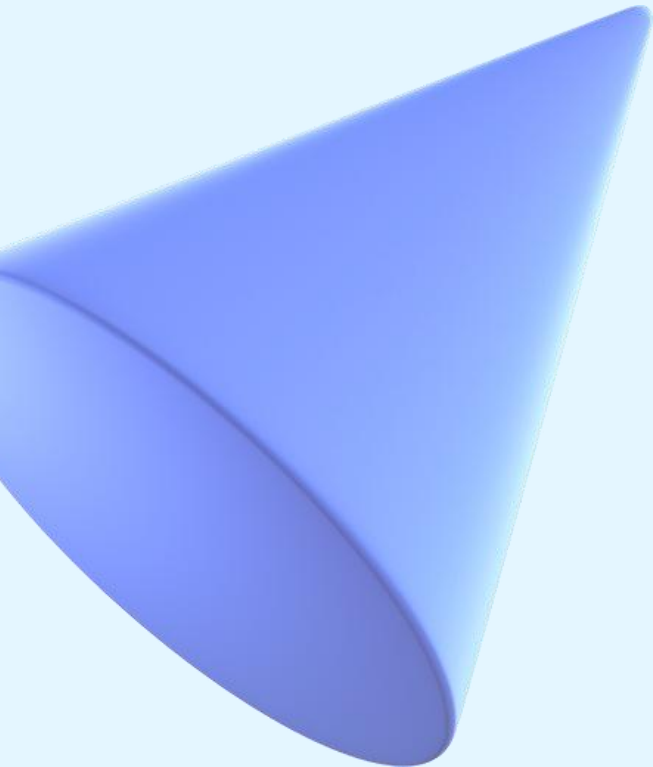
Major Results



70% ↓
Manual Work

80% ⚡
Faster Process

95% ✓
Data Accuracy

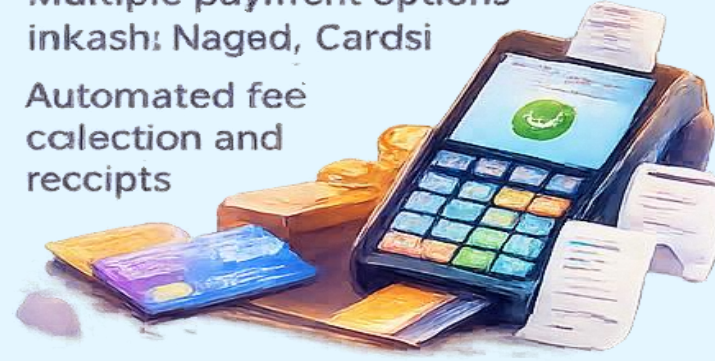
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-  Reduction in Manual Workload: 70% less paperwork
 -  Faster Processing Time: Admission process reduced from weeks to days
 -  Improved Data Accuracy: 95% reduction in data entry errors
 -  Enhanced Communication: Real-time notifications to 1000+ users
 -  Overall System Reliability: 99.5% uptime and availability

Significance of the Study

Online Payment Gateway Integration

Multiple payment options
inkash, Naged, Cards

- ▶ Automated fee collection and receipts



Advanced Analytics Dashboard

Predictive analytics for student performance

- ▶ Business intelligence and reporting tools



Parent and Alumni Portals

Parent access to student progress

- ▶ Alumni networking and engagement platform



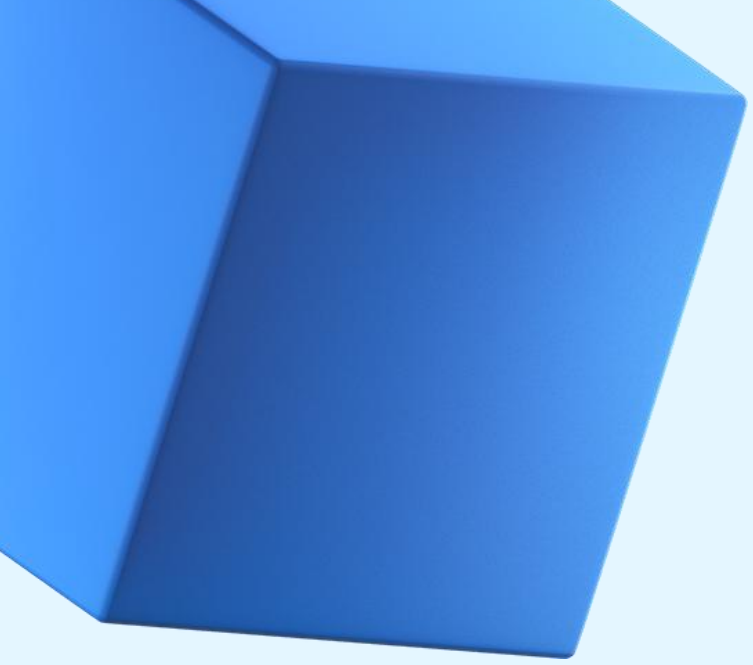
Cloud Scalability Enhancement

- ▶ Multi-region deployment
- ▶ Auto scaling infrastructure



AI-Driven Decision Support

Machine learning for admissions
Automated academic counseling



Conclusion

- ✓ Identified critical problems in traditional university systems
- ✓ Developed comprehensive multi-platform solution
- ✓ Achieved significant operational improvements
- ✓ System successfully deployed and tested

Questions & Answers

